

BVAR vs. Information Criteria: A Monte Carlo Simulation Study

Akın Aydın* Ferdinand Loser†

June 19, 2026

Abstract

This is the abstract for your paper. Briefly state the purpose of the research, the principal results, and major conclusions. The abstract should be self-contained and citation-free.

Keywords: Keyword 1, Keyword 2, Keyword 3

JEL Codes: C15, C32

*Ludwig-Maximilian University of Munich, akin.aydin@campus.lmu.de

†Ludwig-Maximilian University of Munich, ferdinand.loser@campus.lmu.de

1 Introduction

Introduce the problem of lag selection and parameter proliferation in Structural VARs. Highlight the debate between hard lag selection (Information Criteria) and Bayesian methods (Shrinkage and Model Averaging).

2 Simulation Design

This Monte Carlo simulation study follows Ivanov and Kilian (2005) design with several alterations in order to make our extensions feasible and to fit the needs of our analysis. First of all, we only use one monthly DGP based on the empirical VAR study of Kilian and Murphy (2014). Therefore, we do not get estimations for quarterly VAR series or VEC series and additionally do not have additional robustness by including several different series with different macroeconomic indicators and movements. The main reason for this is our tight time constraint. However, we think that the DGP of Kilian and Murphy (2014) can still work as a good benchmark for the inference of how different methods work for monthly VAR series as Ivanov and Kilian (2005) did not find stark differences between different DGPs of the same type (monthly, quarterly, VEC). Our DGP is therefore created by following the empirical VAR study of Kilian and Murphy (2014) and is based on the following 4 variables: percent change in global oil production, real activity index from Kilian(AER 2009), the log real price of oil, and changes in OECD crude oil inventories

2.1 Data Generating Process

Explain the true DGP based on Kilian and Murphy (2014) or other literature.

2.2 Optimal Prior Selection via Marginal Data Density

Describe how the shrinkage parameter (τ) is treated as an optimization problem rather than a fixed heuristic.

3 Simulation Results

Present the results of the Monte Carlo simulations, broken down by the specific strategies employed to handle model uncertainty.

3.1 Behavior of Standard Information Criteria

Before evaluating the Bayesian estimators, we examine the baseline lag detection performance of the classical Information Criteria.

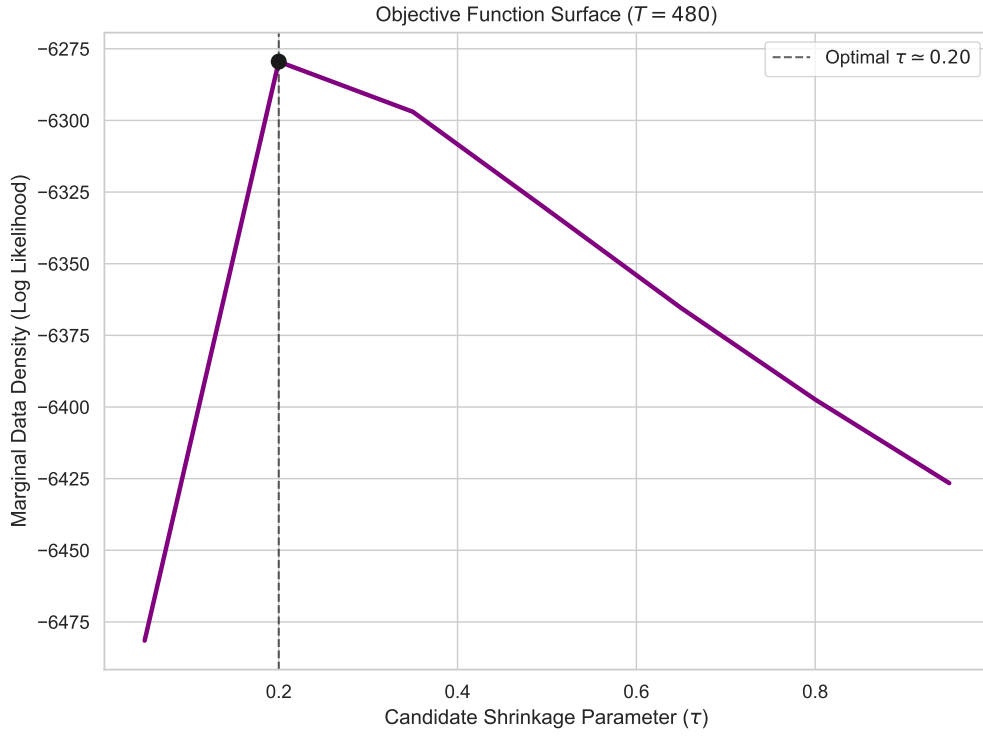


Figure 1: Objective Function Surface for Marginal Data Density Optimization. The peak represents the optimal balance between model fit and parameter penalty.

Table 1: Average Evaluated Lag Order ($\hat{p}$) selected by Information Criteria.

p_0	Estimator	AIC	BIC	HQC
	T			
4	96	9.37	1.25	1.85
	144	2.70	1.38	2.01
	240	3.07	1.97	2.07
	480	3.62	2.00	2.38
6	96	10.55	1.18	1.78
	144	3.09	1.54	2.02
	240	3.92	1.96	2.07
	480	5.60	2.02	2.60
8	96	11.27	1.19	1.99
	144	4.97	1.61	2.00
	240	6.46	1.95	2.07
	480	8.02	2.00	2.58
10	96	11.16	1.24	2.13
	144	4.83	1.61	2.01
	240	6.40	1.97	2.13
	480	8.97	2.00	2.70

3.2 The Shrinkage Strategy: Priors vs. Data

This section compares hard lag selection against dynamic BVARs. We analyze both the bias-variance tradeoff and how the optimal prior relaxes asymptotically as sample sizes increase.

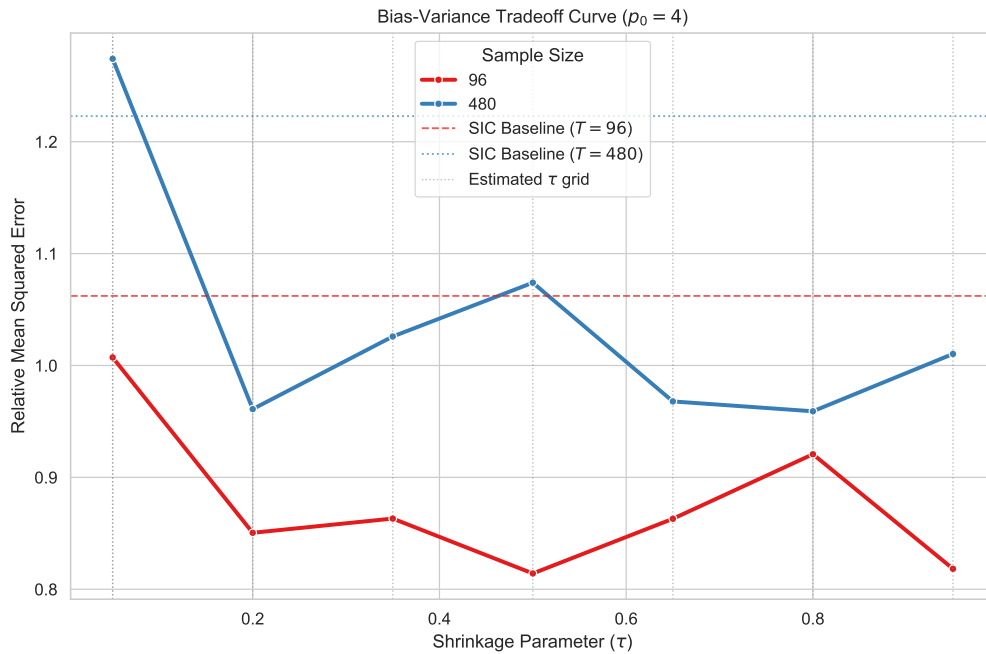


Figure 2: Comparison of Bias-Variance Tradeoff: BVAR vs. Hard Selection. The curve illustrates how the optimal τ balances bias and variance, achieving superior MSE performance compared to fixed lag selection.

Table 2: Relative Mean Squared Error: Hard Selection vs. Bayesian Shrinkage.

T	p_0	AIC	BIC	HQC	BVAR ($\tau = 0.05$)	BVAR ($\tau = 0.20$)	BVAR ($\tau = 0.35$)	BVAR ($\tau = 0.50$)	BVAR ($\tau = 0.65$)	BVAR ($\tau = 0.80$)	BVAR ($\tau = 0.95$)	BVAR RW ($\tau = 0.20$)	BVAR RW ($\tau = 0.50$)	N
96	4	1.518	1.062	1.071	1.057	0.884	0.804	0.847	0.871	0.822	0.799	1.514	0.914	
	6	1.420	1.117	1.066	0.941	0.849	0.763	0.818	0.743	0.780	0.859	1.386	0.779	
	8	1.189	0.953	1.009	1.008	0.806	0.763	0.803	0.781	0.770	0.818	1.427	0.800	
	10	1.063	0.874	0.774	0.821	0.733	0.627	0.715	0.657	0.601	0.593	1.089	0.725	
144	4	1.100	1.160	1.124	1.391	1.154	1.100	1.005	0.922	1.070	0.975	1.644	1.111	
	6	1.033	0.948	1.056	1.037	0.857	0.816	0.757	0.766	0.812	0.763	1.146	0.805	
	8	0.845	0.947	0.918	1.060	0.785	0.769	0.771	0.836	0.798	0.799	1.180	0.794	
	10	0.882	0.938	0.923	0.901	0.831	0.818	0.809	0.767	0.829	0.788	1.071	0.843	
240	4	1.042	1.040	1.080	1.299	0.982	0.976	0.978	0.972	0.916	0.896	1.303	0.968	
	6	1.076	0.981	0.963	1.247	1.046	1.019	0.933	0.887	1.010	0.847	1.318	0.949	
	8	0.971	0.869	0.890	1.162	0.852	0.962	0.864	0.944	0.883	0.871	1.073	0.900	
	10	0.926	0.952	0.973	1.051	0.813	0.922	0.882	0.928	0.895	0.947	0.955	0.935	
480	4	1.095	1.223	1.218	1.330	1.170	1.032	1.023	0.937	1.039	0.917	1.265	1.137	
	6	1.040	1.131	1.101	1.392	1.247	1.163	1.147	1.088	1.033	1.197	1.349	1.239	
	8	0.992	0.982	1.078	1.108	1.015	0.949	0.794	0.892	0.869	0.899	0.941	0.923	
	10	0.905	1.082	0.924	1.168	0.796	0.971	0.901	0.862	0.918	0.907	1.004	0.953	

Table 3: Ratio of Relative MSE: Estimating with True Lag Order (p_0) vs. Maximum Lag Order ($p_{\max}$). Values < 1.0 indicate p_0 outperforms $p_{\max}$.

T	p_0	τ	τ	τ	τ	τ	τ	τ	MDD
		0.05	0.20	0.35	0.50	0.65	0.80	0.95	τ
96	4	0.946	1.102	1.104	1.016	0.925	1.002	1.026	1.115
	6	1.172	1.074	1.168	1.046	1.115	1.088	0.972	1.046
	8	0.998	1.142	0.989	1.081	1.073	1.102	1.069	1.133
	10	0.801	0.933	1.008	0.907	0.943	1.064	0.964	0.918
144	4	0.813	0.845	0.977	1.010	1.129	0.912	1.149	0.845
	6	0.996	0.980	1.024	0.990	1.001	0.948	1.023	0.980
	8	0.938	1.029	1.026	1.064	0.943	1.109	1.031	1.029
	10	0.995	0.891	0.872	0.875	0.914	0.827	0.784	0.891
240	4	1.022	1.061	1.030	1.031	0.957	1.169	1.223	1.061
	6	0.921	0.881	0.892	0.918	0.945	0.837	1.160	0.881
	8	1.001	1.084	0.922	0.946	1.004	1.027	1.043	1.084
	10	1.058	1.135	0.970	1.000	1.006	0.969	0.938	1.135
480	4	1.118	1.139	1.213	1.122	1.410	1.132	1.196	1.074
	6	0.935	0.862	1.006	0.848	0.909	0.960	0.844	1.021
	8	1.045	0.928	0.927	1.138	1.046	0.941	1.017	0.862
	10	0.950	1.313	0.874	1.088	1.043	1.039	0.987	1.071

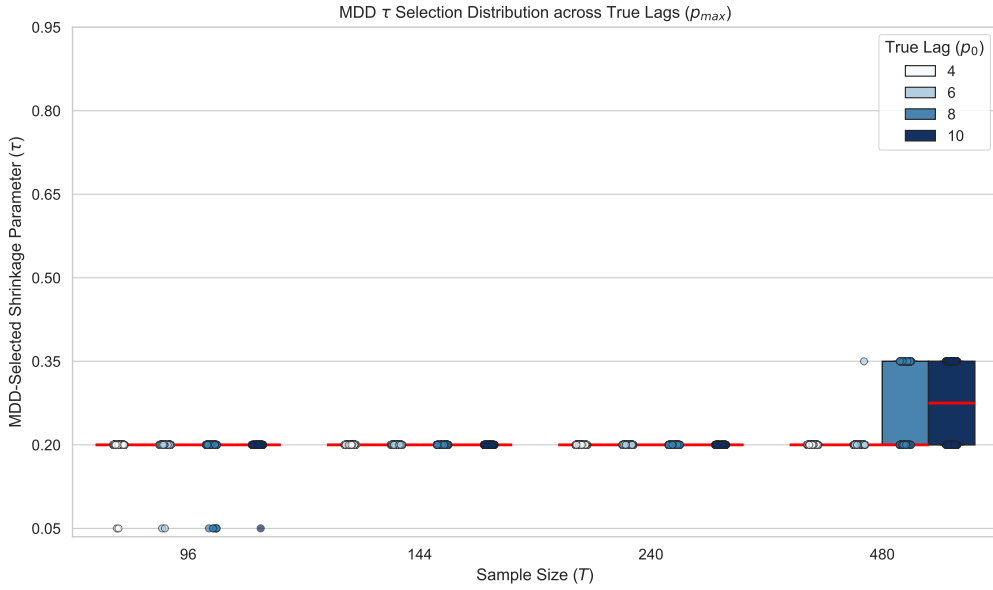


Figure 3: The MDD selected priors for different sample sizes for $p = p_{\max} = 12$, $p = p_0 = 4$.

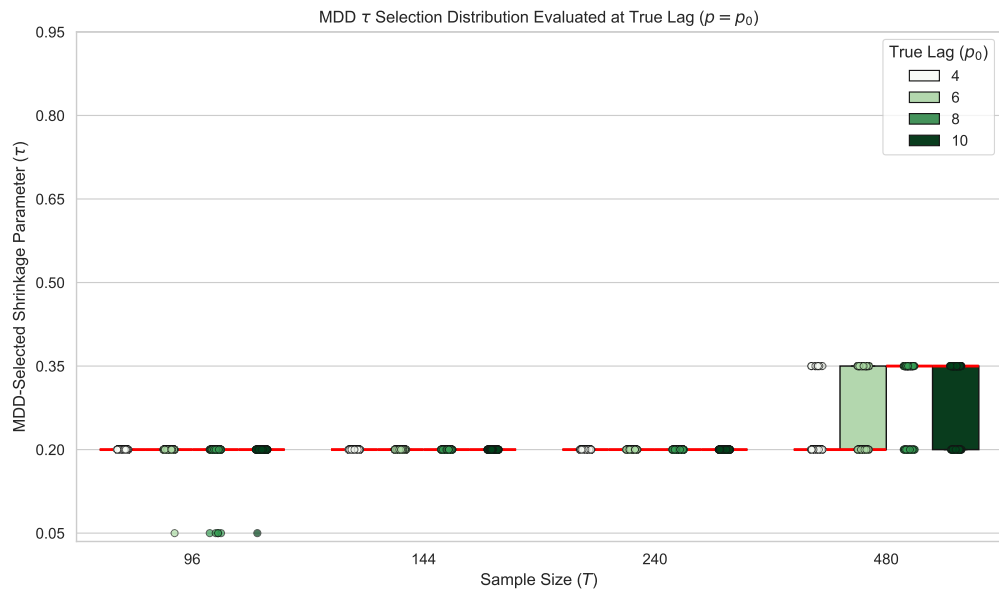


Figure 4: The MDD selected priors for different sample sizes for $p = p_0 = 12$.

4 Conclusion

Summarize the main findings, specifically highlighting how optimal τ selection and true BMA systematically outperform standard econometric heuristics in finite samples.

References

- Ivanov, Ventzislav and Lutz Kilian (Jan. 14, 2005). “A Practitioner’s Guide to Lag Order Selection For VAR Impulse Response Analysis”. In: *Studies in Nonlinear Dynamics & Econometrics* 9.1. ISSN: 1558-3708. DOI: [10.2202/1558-3708.1219](https://doi.org/10.2202/1558-3708.1219). URL: <https://www.degruyter.com/document/doi/10.2202/1558-3708.1219/html> (visited on 04/22/2026).
- Kilian, Lutz and Daniel P. Murphy (Apr. 1, 2014). “The Role of Inventories and Speculative Trading in the Global Market for Crude Oil”. In: *Journal of Applied Econometrics* 29.3, pp. 454–478. ISSN: 1099-1255. DOI: [10.1002/jae.2322](https://doi.org/10.1002/jae.2322). URL: <https://onlinelibrary.wiley.com/doi/10.1002/jae.2322> (visited on 05/12/2026).